SIMATS ENGINEERING

CSA6303- Threat Intelligence and Network Security

A1-Malware Analysis Assignment

Assessment Questions

Subject	Threat Intelligence	**Total Marks**	50
Questions	10 (all compulsory)	**Marks per Question**	5
Reg.No	192425380	**Name**	T. Bhanu Pavan Varma
Faculty	Dr.V.Parthipan	**Slot**	A

Instructions: Answer all questions. Each question carries 5 marks. Support answers with examples and diagrams where appropriate.

CO3: Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

Code of Conduct:

I T. Bhanu Pavan Varma(192425380) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of Student

Questions :

1. Malware Analysis Using Static Techniques:A security analyst receives a suspicious executable file attached to a phishing email. Before running the file, the analyst wants to examine its properties, file hash, strings, and imported functions. Explain how static malware analysis can help identify suspicious characteristics and assess the potential risk without executing the file.

2. Malware Analysis Using Dynamic Techniques: An organization suspects that an employee has downloaded a malicious program that may modify system files and communicate with an external server. Explain how dynamic malware analysis can be performed in an isolated virtual environment to observe the program's behavior and identify potential Indicators of Compromise (IoCs).

3. Malware Investigation Using a Threat Intelligence Platform: A company's security team discovers a suspicious file on an employee's computer. The file hash appears in a threat intelligence platform associated with a known malware family. Explain how the analyst can use threat intelligence sources to investigate the file, assess the reliability of the information, and recommend suitable containment measures.

4. Malware Detection Through Data Collection and Correlation: A university detects unusual processes on several computers, suspicious DNS requests, and unexpected outbound network connections. Explain how an analyst can collect and correlate endpoint logs, DNS records, firewall logs, and file hashes to determine whether the computers are infected with malware.

5. Malware Investigation Using SIEM Tools:A financial institution's Security Information and Event Management (SIEM) system generates alerts about suspicious process execution, repeated failed logins, and unusual network connections from the same workstation. Describe how the security analyst can use SIEM capabilities to correlate these events, investigate the possible malware infection, and prioritize the incident.

6. Malware Detection and Log Analysis Using Splunk: A company uses Splunk to monitor its network and endpoint logs. The security team suspects that malware is attempting to contact an external command-and-control server. Explain how Splunk can be used to search relevant logs, identify suspicious communication patterns, correlate events, and document evidence of possible malware activity.

7. Malware Investigation Using the ELK Stack: An e-commerce company uses the ELK Stack (Elasticsearch, Logstash, and Kibana) to monitor web server logs, firewall records, and endpoint events. The security team notices repeated connections to suspicious domains and unusual file activity. Explain how the ELK Stack can support malware investigation through log collection, processing, searching, visualization, and correlation.

8. Threat Hunting for Malware Activities: An organization's security team suspects that malware has infected an employee workstation but has not received any antivirus alerts. The workstation shows unusual CPU usage, unfamiliar processes, and connections to unknown external IP addresses. Describe a threat-hunting approach to investigate the workstation, identify suspicious behaviors, collect relevant IoCs, and recommend remediation measures.

9. Comparing Malware Analysis Tools Used in the IT Industry:A cybersecurity team needs to investigate a suspicious file and determine whether it contains malicious code. Compare static analysis tools, dynamic analysis tools, network traffic analyzers, and threat intelligence platforms in terms of their purpose and usefulness. Recommend an appropriate combination of tools for conducting a malware investigation.

10. Malware Analysis Report and Incident Response: A company discovers a suspicious executable on an employee's computer. Initial analysis identifies an unfamiliar process, a modified system configuration, and communication with a suspicious external domain. Prepare an outline for a malware analysis report that includes file information, analysis methodology, observed behavior, IoCs, threat intelligence findings, risk assessment, and recommended containment and recovery actions.

Q1. Malware Analysis Using Static Techniques

Definition: Static malware analysis is the examination of a suspicious file without executing it. It helps identify malicious characteristics while reducing the risk of infection.

Techniques used

1. File Properties: Examine the file name, size, type, digital signature, and metadata to identify suspicious characteristics.

2. File Hashing: Generate MD5 or SHA-256 hashes to uniquely identify the file and compare it with known malware databases.

3. Strings Analysis: Extract readable text, URLs, IP addresses, file paths, and suspicious commands embedded in the executable.

4. Imported Functions: Examine imported functions for suspicious capabilities, such as file modification, process creation, registry changes, or network communication.

5. Header Analysis: Inspect executable headers and sections using tools such as PEStudio or Detect It Easy to identify unusual structures.

6. Threat Intelligence Lookup: Compare file hashes and extracted indicators with trusted threat intelligence sources.

Tools: PEStudio, Detect It Easy (DIE), strings, and VirusTotal.

Conclusion: Static analysis helps identify potential malware, understand its capabilities, and assess risk without executing the suspicious file. However, obfuscation and encryption may hide malicious behavior.

Q2. Malware Analysis Using Dynamic Techniques

Definition: Dynamic malware analysis involves executing a suspicious program in a controlled, isolated environment to observe its actual behavior.

Procedure

1. Prepare a Virtual Machine: Create an isolated virtual environment with a clean operating system and a snapshot for recovery.
2. Establish a Baseline: Record existing processes, files, registry settings, and network activity before execution.
3. Execute the Sample Safely: Run the suspicious program inside the isolated environment while monitoring its behavior.
4. Monitor File and System Changes: Observe file creation, deletion, registry modifications, and changes to system settings.
5. Monitor Processes: Record process creation, child processes, command-line arguments, and unusual resource usage.
6. Capture Network Traffic: Use Wireshark or an isolated network simulator to identify DNS requests, external connections, and suspicious communication
7. Collect IoCs: Record file hashes, suspicious domains, IP addresses, file paths, and registry keys.
8. Restore the Environment: Revert the virtual machine to its clean snapshot after analysis.

Tools used

- Cuckoo Sandbox: Automates behavioral malware analysis.
- Process Monitor (Procmon): Records file, registry, and process activity.
- Wireshark: Captures and analyzes network traffic.

Conclusion: Dynamic analysis reveals runtime behavior and helps identify Indicators of Compromise (IoCs). Samples should be analyzed only in a properly isolated lab, because malware can detect virtual machines or attempt to escape its environment.

Q3. Malware Investigation Using a Threat Intelligence Platform

Definition: A threat intelligence platform collects and analyzes threat information to help security teams identify malicious files, infrastructure, and attacker behavior.

Investigation procedure

1. Identify the File: Collect the suspicious file's name, location, size, and SHA-256 hash.
2. Search Threat Intelligence Sources: Submit the hash to services such as VirusTotal or other trusted threat intelligence platforms.
3. Examine Reputation: Review malware classifications, detection results, related domains, IP addresses, and reported malware families.
4. Validate the Findings: Compare multiple independent sources, check the dates of reports, and consider whether the detections agree.
5. Analyze File Behavior: Use static or dynamic analysis to verify the reported capabilities and identify additional indicators.
6. Assess the Risk: Determine whether the file is running, which systems are affected, and whether sensitive data may be exposed.
7. Contain the Threat: Isolate the affected computer when necessary, block confirmed malicious indicators, and disable or secure compromised accounts.
8. Document Findings: Record evidence, confidence levels, affected assets, and recommended remediation.

Assessing reliability

- Multiple independent detections increase confidence but do not guarantee maliciousness.
- False positives may occur, especially for legitimate administrative tools.
- Old intelligence may contain outdated IP addresses or domains.
- A hash match identifies the same file content, but does not by itself prove that the file executed on the affected system.

Conclusion: Threat intelligence helps analysts identify known malware faster, validate suspicious files, understand associated threats, and make informed containment decisions.

Q4. Malware Detection Through Data Collection and Correlation

Definition: Data correlation is the process of connecting security events from different sources to identify patterns that may indicate malware infection.

Investigation steps

1. Collect Endpoint Logs: Gather process execution records, file modifications, antivirus alerts, and system events from affected computers.
2. Collect DNS Records: Identify suspicious domain lookups and repeated requests to unfamiliar destinations.
3. Analyze Firewall Logs: Examine outbound connections, destination IP addresses, ports, protocols, and transferred data volumes.
4. Calculate File Hashes: Generate SHA-256 hashes for suspicious executables and compare them with threat intelligence sources.
5. Normalize Timestamps: Align events from different systems using consistent time zones and synchronized clocks.
6. Correlate Events: Connect process activity, DNS requests, and outbound connections by timestamp, hostname, user, and destination.
7. Identify Patterns: Look for repeated connections, unusual processes, suspicious persistence, or similar file hashes across several computers.

Example

Data source	Observation	Possible significance
Endpoint logs	Unknown executable starts	Possible malware execution
DNS logs	Executable's host repeatedly queries a suspicious domain	Possible command-and-control lookup
Firewall logs	Outbound connections to the resolved IP	Possible external communication
File hash	Matches a known malware sample	Strong supporting evidence

Conclusion: Correlating endpoint, DNS, firewall, and file-hash information helps analysts establish a timeline, identify infected computers, and distinguish suspicious behavior from normal activity.

Q5. Malware Investigation Using SIEM Tools

Definition: Security Information and Event Management (SIEM) tools collect and correlate logs from multiple sources to detect, investigate, and prioritize security incidents.

How SIEM supports investigation

1. Centralized Log Collection: Collect process, authentication, firewall, DNS, and endpoint security events.
2. Event Correlation: Link suspicious process execution with repeated failed logins and unusual network connections from the same workstation.
3. Alert Analysis: Examine event timestamps, affected accounts, hostnames, source and destination IP addresses, and alert descriptions.
4. Threat Intelligence Integration: Compare domains, IP addresses, and file hashes against trusted intelligence feeds.
5. Timeline Reconstruction: Establish the sequence of login attempts, process execution, and external communication.
6. Scope Identification: Search for similar events on other endpoints to determine whether the threat has spread.
7. Risk Prioritization: Evaluate evidence confidence, active exploitation, system importance, and potential data exposure.
8. Incident Response: Escalate confirmed threats, isolate affected endpoints, and secure compromised accounts.
9. Evidence Preservation: Retain relevant logs, alert details, and investigation notes.
10. Reporting: Document the findings, containment actions, and lessons learned.

Example SIEM tools

- Splunk Enterprise Security
- Elastic Security
- Microsoft Sentinel

Conclusion: SIEM correlation helps identify relationships between apparently unrelated events, reduce investigation time, and prioritize malware incidents according to their risk.

Q6. Malware Detection and Log Analysis Using Splunk

Definition: Splunk is a platform that searches, analyzes, and visualizes machine-generated data, including security logs, to help investigate suspicious activity.

Investigation procedure

1. Collect Logs: Ingest endpoint, DNS, firewall, proxy, and authentication logs into Splunk.
2. Search Network Connections: Identify repeated outbound connections to unfamiliar IP addresses or domains.
3. Investigate DNS Activity: Search for repeated DNS requests, unusual domains, and suspicious resolution patterns.
4. Analyze Process Execution: Search endpoint logs for unfamiliar executable names, unusual command lines, and unexpected parent-child process relationships.
5. Correlate Events: Connect process execution with DNS queries and outbound network connections using timestamps and hostnames.
6. Check Threat Intelligence: Compare suspicious IPs, domains, and file hashes against trusted threat intelligence sources.
7. Prioritize Alerts: Consider the severity, asset importance, event frequency, and confidence of the evidence.
8. Preserve Evidence: Export relevant search results and record timestamps, source logs, and investigation notes.

Example Splunk Search Processing Language (SPL)

The following example assumes network logs contain fields named dest_ip and bytes_out:

```
index=network earliest=-24h
| stats count sum(bytes_out) AS total_bytes
   BY src_ip dest_ip
| sort - total_bytes
```

This query summarizes outbound network connections by source and destination IP address and ranks them by transferred bytes. Large transfers are leads for investigation, not proof of malware.

Conclusion: Splunk helps analysts identify unusual communication patterns, correlate evidence across data sources, and document potential command-and-control activity.

Q7. Malware Investigation Using the ELK Stack

Definition: The ELK Stack consists of Elasticsearch, Logstash, and Kibana. It supports centralized log collection, processing, searching, analysis, and visualization.

Components and their roles

Component	Role
Elasticsearch	Stores and searches indexed security logs.
Logstash	Collects, parses, transforms, and enriches log data.
Kibana	Provides dashboards, visualizations, searches, and investigation views.
Beats / Elastic Agent	Collects logs and endpoint telemetry from monitored systems.

Malware investigation process

1. Log Collection: Collect web server, firewall, DNS, and endpoint events using Elastic Agent or other supported collectors.
2. Log Processing: Parse timestamps, normalize field names, and enrich events with host and threat intelligence information.
3. Data Indexing: Store processed events in Elasticsearch for efficient searching.
4. Search and Filtering: Search for suspicious domains, unfamiliar processes, file changes, and repeated outbound connections.
5. Visualization: Use Kibana dashboards to identify spikes in network activity and suspicious behavior across hosts.
6. Event Correlation: Relate endpoint processes to DNS requests and firewall connections using host identifiers and timestamps.
7. Threat Intelligence Comparison: Compare file hashes and network indicators with trusted threat intelligence sources.
8. Incident Response: Prioritize affected endpoints, preserve evidence, and coordinate containment and remediation.

Conclusion: The ELK Stack enables security teams to analyze large volumes of logs, identify suspicious patterns, correlate events, and investigate possible malware infections.

Q8. Threat Hunting for Malware Activities

Definition: Threat hunting is a proactive investigation process used to identify hidden malicious activity that automated security tools may have missed.

Threat-hunting approach

1. Form a Hypothesis: Suspect that an unfamiliar process is generating unusual CPU usage and communicating with an external attacker-controlled server.
2. Collect Endpoint Data: Gather process details, command-line arguments, file paths, startup entries, and relevant system logs.
3. Inspect Running Processes: Identify unusual process names, unsigned executables, suspicious parent-child relationships, and unexpected resource consumption.
4. Analyze Network Activity: Review active connections, destination IPs, DNS queries, and traffic patterns using endpoint telemetry or Zeek/Wireshark.
5. Check File Reputation: Calculate file hashes and compare them with trusted threat intelligence databases.
6. Search for Persistence: Examine scheduled tasks, startup entries, and other authorized investigation points for suspicious modifications.
7. Correlate Evidence: Match process activity with DNS queries, external connections, and system changes.
8. Collect IoCs: Record malicious file hashes, suspicious domains, IP addresses, file paths, and relevant registry keys.

Recommended tools

- Microsoft Defender for Endpoint: Endpoint investigation and isolation.
- Sysmon: Detailed process, network, and system event logging.
- Wireshark: Network packet analysis.
- VirusTotal: File and indicator reputation checks.
- Splunk or Elastic Security: Log correlation and incident investigation.

Conclusion: Proactive hunting can reveal infections that evade antivirus alerts by examining behavior, correlating endpoint and network evidence, and responding to verified threats.

Q9. Comparing Malware Analysis Tools Used in the IT Industry

Different tools serve different purposes. Combining them provides a more complete picture of a suspicious file.

Tool category	Examples	Main purpose	Usefulness
Static analysis	PEStudio, Detect It Easy, strings	Examine file structure, metadata, strings, imports, and hashes without execution.	Identifies suspicious properties and potential capabilities.
Dynamic analysis	Cuckoo Sandbox, Procmon	Observe runtime behavior in a controlled environment.	Reveals file changes, process activity, persistence, and network behavior.
Network analysis	Wireshark, Zeek	Examine packets and network connection records.	Identifies suspicious DNS queries, external communication, and possible command-and-control traffic.
Threat intelligence	VirusTotal, AlienVault OTX	Check indicators against known threat information.	Helps identify known malware and related infrastructure.
SIEM and log analysis	Splunk, Elastic Security	Correlate endpoint, authentication, and network events.	Helps reconstruct incidents and identify affected systems.

Recommended combination

1. PEStudio and SHA-256 hashing: Begin with static inspection and file identification.
2. VirusTotal: Check the file hash against known threat intelligence.
3. Cuckoo Sandbox or Procmon: Observe behavior in an isolated environment if additional evidence is required.
4. Wireshark or Zeek: Investigate network communication and extract network IoCs.
5. Splunk or Elastic Security: Correlate the findings with organizational logs and identify other potentially affected systems.

Conclusion: Static analysis identifies suspicious file characteristics, dynamic analysis reveals behavior, network analysis investigates communications, and threat intelligence provides context. SIEM tools connect these findings to the wider incident.

Q10. Malware Analysis Report and Incident Response

A malware analysis report documents the suspicious file, investigation methodology, evidence, findings, risk, and recommended corrective actions.

Sample report outline

Malware Analysis Report

Incident type: Suspected malware infection

1. Executive Summary

A suspicious executable was discovered on an employee workstation. Initial observations include an unfamiliar process, a modified system configuration, and communication with a suspicious external domain.

2. File Information

- File name and path
- File size and type
- SHA-256 hash
- Digital signature and metadata
- Detection date and affected host

3. Analysis Methodology

- Static analysis: file properties, strings, imports, and hashes.
- Dynamic analysis: process behavior, file changes, and network activity in an isolated environment.
- Threat intelligence: reputation checks for hashes, domains, and IP addresses.
- Log correlation: endpoint, DNS, firewall, and authentication records.

4. Observed Behavior

- An unfamiliar process executed.
- A system configuration was modified.
- An external domain was contacted.

- Further investigation is required to establish persistence, the scope of infection, and possible data exposure.

5. Indicators of Compromise (IoCs)

Indicator	Example
File hash	SHA-256 hash of the suspicious executable
File path	Location of the suspicious executable
Domain	Observed external domain
IP address	Resolved destination IP
Process	Suspicious process name and execution details
System change	Modified registry key or configuration file

Replace these descriptions with verified incident evidence. Do not invent indicator values.

6. Threat Intelligence Findings

Record reputation results, related malware reports, known attacker techniques, source reliability, and confidence in each finding.

7. Risk Assessment

Assess the likelihood of continued compromise, affected systems, potential data exposure, privileges obtained, and business impact.

8. Containment and Eradication

- Isolate the affected workstation when appropriate.
- Block verified malicious indicators.
- Disable or secure compromised accounts and revoke active sessions.
- Preserve forensic evidence.
- Remove malicious artifacts and remediate the initial entry point.

9. Recovery and Prevention

- Restore systems from a trusted state if required.
- Patch vulnerabilities and strengthen endpoint controls.
- Reset affected credentials and review access permissions.

- Add appropriate SIEM/EDR detection rules and monitor for recurrence.

10. Conclusion and Recommendations

Summarize confirmed findings, unresolved questions, affected assets, remediation priorities, and follow-up monitoring requirements.

Conclusion: A structured malware analysis report helps security teams communicate findings, prioritize response actions, preserve evidence, and improve defenses against future malware incidents.